SPEED DETECTOR

Project Based Learning (PBL) Report

for the course

Internet of Things (20CS32001)

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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A.Y 2025-26 (Sem-II)

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CERTIFICATE

This is to certify that the Project-Based Learning (PBL) report entitled **"SPEED DETECTOR"** has been carried out by **A.Sharan Kumar Raju (Roll No. 23R11A05U8), D.Romith Kumar (Roll No. 23R11A05V4), K.Advaith (Roll No. 23R11A05W7),** of III Year II Semester, B.Tech in Computer Science and Engineering, during the academic year 2025-2026, under the guidance of **T. Neelima , Assistant Professor**, Department of Computer Science and Engineering. This work is submitted in partial fulfillment of the requirements for the Project-Based Learning (PBL) component of the **Internet of Things Lab (20CS32L01).**

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ACKNOWLEDGEMENT

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DECLARATION BY THE CANDIDATE

We hereby declare that the Project-Based Learning (PBL) report entitled **"SPEED DETECTOR"** submitted to the Department of Computer Science and Engineering, Geethanjali College of Engineering and Technology, is a record of original work carried out by us during the academic year 2025-2026, under the guidance of **T.Neelima , Assistant Professor,** Department of Computer Science and Engineering. We further declare that this report has not been submitted to any other institution or university for the award of any degree.

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Date: 18/03/2026

TABLE OF CONTENTS

S.No.	Contents	Page No.
1.	Abstract	5
2.	Introduction	6 - 7
3.	System Design	8 - 9
4.	Implementation	10 - 19
5.	Conclusion	20
6.	References	21

1. Abstract

Speed measurement plays a crucial role in traffic management and safety systems. This project presents the design and implementation of a Speed Detection System using an Arduino Uno microcontroller and infrared (IR) sensors. Two IR sensors are placed at a fixed distance to detect the movement of a vehicle or object. When the object passes the first sensor, the timer starts, and when it reaches the second sensor, the timer stops. Using the time difference and known distance, the system calculates the speed based on the distance–time relationship.

The calculated speed is displayed in real time on a 16×2 LCD display using an I2C interface, providing a simple user interface. Additionally, a buzzer is incorporated to indicate overspeeding when the measured speed exceeds a predefined limit. The system is cost-effective, easy to implement, and demonstrates the practical application of embedded systems, sensor interfacing, and real-time processing.

2. Introduction

2.1 About the project

The **Speed Detector System using Arduino and IR Sensors** is designed to measure the speed of a moving vehicle or object by detecting the time taken to travel between two points. The system consists of two IR sensors placed at a fixed distance, an Arduino UNO microcontroller, an LCD display with an I2C interface, and a buzzer for alert indication.

When a vehicle passes the first IR sensor, the Arduino records the start time. As the vehicle continues to move and crosses the second sensor, the Arduino records the end time. Using the formula: Speed = Distance / Time

The system calculates the speed of the vehicle. The calculated speed is then converted into kilometers per hour (km/h) and displayed on the LCD screen.

If the detected speed exceeds the specified speed limit, the system activates a buzzer to alert over speeding. This system demonstrates a basic real-time speed detection mechanism and can be used for educational purposes to understand embedded systems, sensors, and microcontroller programming.

2.2 Project Objectives

The main objectives of this project are:

- To design and implement a vehicle speed detection system using Arduino.
- To measure the speed of a moving object using two infrared sensors.
- To calculate speed using the distance-time relationship.
- To display the measured speed on a 16×2 LCD display.
- To provide an alert using a buzzer when the speed exceeds a predefined limit.
- To demonstrate the use of microcontrollers and sensors in traffic monitoring applications.

2.3 Project Outcomes

After successful completion of this project:

- A working prototype of a speed detection system is developed.
- The system can measure the speed of a moving vehicle using IR sensors.
- The calculated speed is displayed clearly on the LCD screen.
- The buzzer alert helps indicate speeding situations.
- The project demonstrates the practical implementation of embedded systems and sensor integration.
- Students gain hands-on experience in Arduino programming, circuit design, and real-time data processing.

3. System Design

3.1 System Architecture

The system architecture represents the overall arrangement and interaction of hardware components in the project.

The architecture consists of the following main components:

i. **IR Sensor 1 and IR Sensor 2**

   These sensors detect the presence of a vehicle. When a vehicle passes in front of the sensors, they send signals to the Arduino.

ii. **Arduino UNO Microcontroller**

   The Arduino acts as the central processing unit of the system. It receives signals from the sensors, measures the time difference between detections, calculates the speed, and controls the LCD display and buzzer.

iii. **LCD Display (16×2 with I2C Module)**

   The LCD displays the calculated speed of the vehicle and system messages such as "No Vehicle Detected."

iv. **Buzzer**

   The buzzer provides an audible alert when the detected speed exceeds the predefined speed limit.

v. **Power Supply**

   The system is powered through the Arduino board using a USB connection or external power supply.

System Flow

Vehicle → IR Sensor 1 → Arduino (Start Timer) → IR Sensor 2 → Arduino (Stop Timer) → Speed Calculation → LCD Display → Overspeed Check → Buzzer Alert

3.2 Modules

The project can be divided into the following functional modules:

i. **Sensor Detection Module**

This module consists of two IR sensors placed at a fixed distance from each other. When a vehicle passes the first sensor, the Arduino records the start time. When the vehicle reaches the second sensor, the Arduino records the end time.

ii. **Time Measurement Module**

The Arduino measures the time difference between the two sensor detections using the internal timing function. This time difference represents how long the vehicle took to travel the distance between the sensors.

iii. **Speed Calculation Module**

Using the measured time and the known distance between the sensors, the Arduino calculates the speed of the vehicle using the formula:Speed = Distance / TimeThe speed is then converted from meters per second to kilometers per hour.

iv. **Display Module**

The LCD display module shows the calculated speed of the vehicle. It also displays system messages such as “Speed Detector” and “No Vehicle Detected.”

v. **Alert Module**

This module activates the buzzer when the detected speed exceeds the predefined speed limit. The buzzer acts as a warning indicator for overspeeding vehicles.

4. Implementation

4.1 Modules Implementation

The implementation of the system is divided into several modules based on their functionality.

i. **Sensor Detection Module Implementation**

Two infrared sensors are used to detect the movement of a vehicle. These sensors are connected to the analog pins A0 and A1 of the Arduino UNO. When a vehicle interrupts the IR beam, the sensor output changes state, which is detected by the Arduino.

The first sensor starts the timing process, while the second sensor stops the timer. This helps determine the time taken by the vehicle to travel the distance between the sensors.

ii. **Time Measurement Module Implementation**

The Arduino uses its internal timing function millis() to measure time. When the first sensor detects the vehicle, the current time is recorded as the start time. When the second sensor detects the vehicle, the end time is recorded.

The time difference between these two values represents the time taken by the vehicle to travel between the sensors.

Time Taken = End Time − Start Time

iii. **Speed Calculation Module Implementation**

Once the time taken is determined, the Arduino calculates the speed using the formula:

Speed = Distance / Time

The distance between the sensors is predefined in the program. Since the speed is initially calculated in meters per second, it is converted into kilometers per hour using a conversion factor.

Speed (km/h) = Speed (m/s) × 3.6

iv. Display Module Implementation

The LCD display with I2C interface is used to show system messages and the calculated speed. The Arduino communicates with the LCD using the I2C protocol through the SDA and SCL pins.

When the speed is calculated, it is displayed on the LCD screen along with the unit km/h.

Example display message:

Speed: 45 km/h

v. Alert Module Implementation

The buzzer is connected to digital pin 13 of the Arduino. If the calculated speed exceeds the predefined speed limit, the Arduino activates the buzzer for five seconds. This serves as an alert for over speeding. If the vehicle does not reach the second sensor within a certain time period, the system displays the message “No Vehicle Detected” on the LCD.

4.2 Sample Code

iot_pbl.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int sensor1 = A0;
const int sensor2 = A1;
const int buzzer  = 13;

unsigned long startTime = 0;
float distance = 1.0;   // meters between sensors

int firstSensor = 0;     // 0 = none, 1 = sensor1, 2 = sensor2

void setup() {
  pinMode(sensor1, INPUT);
  pinMode(sensor2, INPUT);
  pinMode(buzzer, OUTPUT);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Speed Detector");
  delay(2000);
  lcd.clear();
}

void loop() {

  // Detect first sensor
  if (firstSensor == 0) {
    if (digitalRead(sensor1) == LOW) {
```

iot_pbl.ino

```
// Detect first sensor
if (firstSensor == 0) {
  if (digitalRead(sensor1) == LOW) {
    startTime = millis();
    firstSensor = 1;
  }
  else if (digitalRead(sensor2) == LOW) {
    startTime = millis();
    firstSensor = 2;
  }
}

// If first was sensor1, wait for sensor2
else if (firstSensor == 1) {

  if (digitalRead(sensor2) == LOW) {
    calculateSpeed();
  }

  // Timeout 3 seconds
  else if (millis() - startTime > 3000) {
    showNoVehicle();
  }
}

// If first was sensor2, wait for sensor1
else if (firstSensor == 2) {

  if (digitalRead(sensor1) == LOW) {
    calculateSpeed();
  }

  // Timeout 3 seconds
```

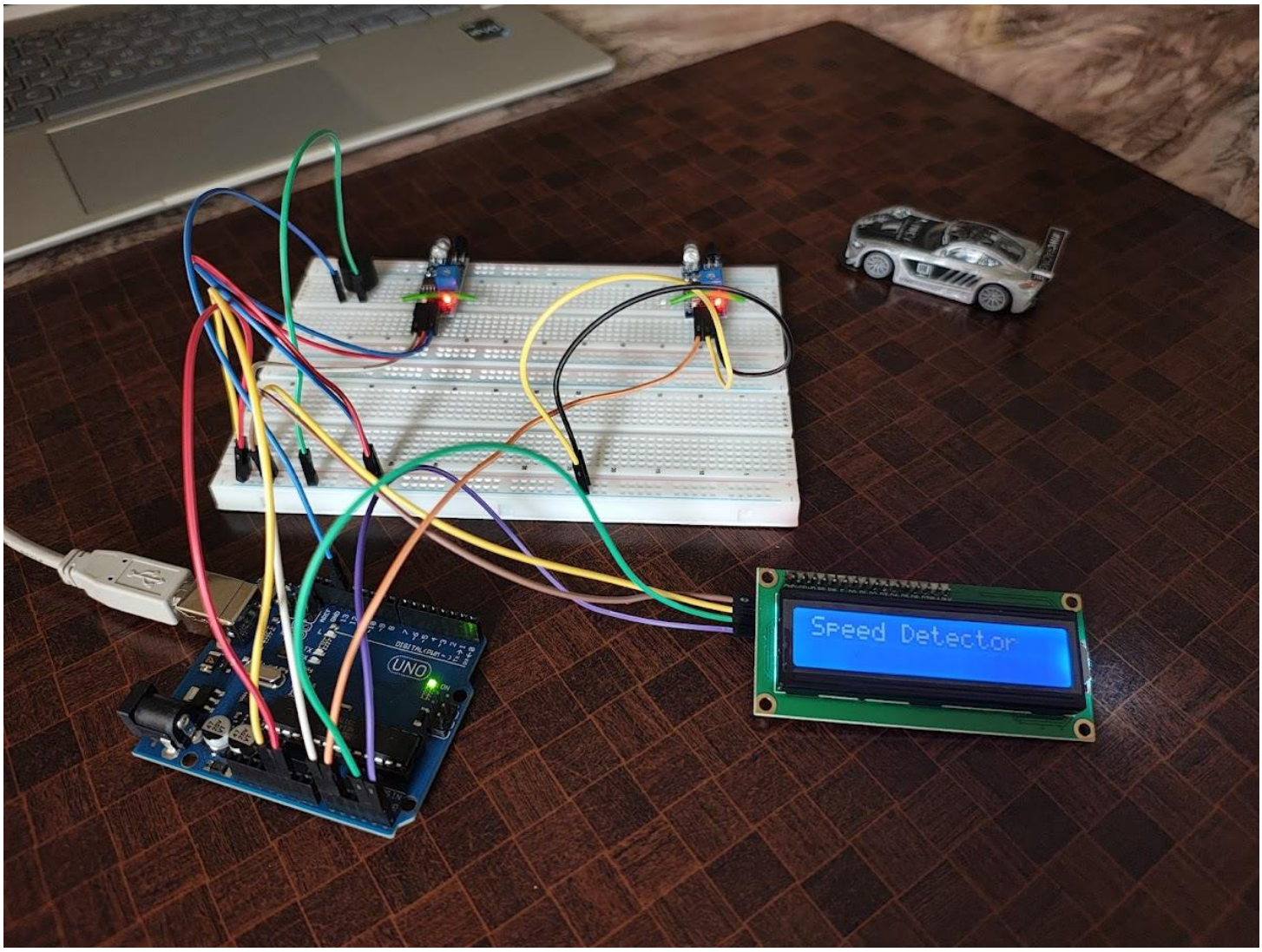
iot_pbl.ino

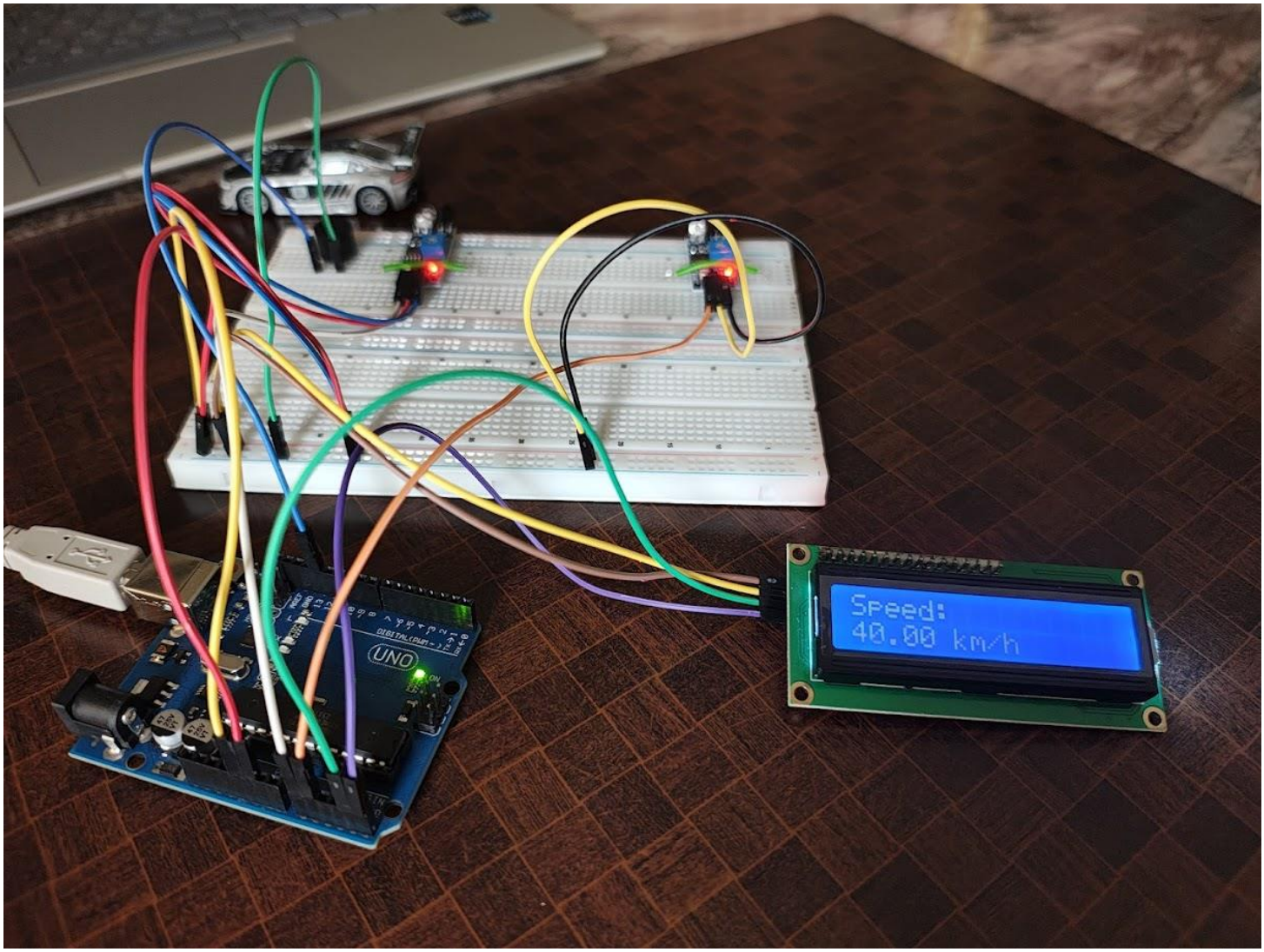
```
    // Timeout 3 seconds
    else if (millis() - startTime > 3000) {
      showNoVehicle();
    }
  }
}

void calculateSpeed() {

  unsigned long endTime = millis();
  unsigned long timeTaken = endTime - startTime;

  if (timeTaken > 0) {

    float speed = distance / (timeTaken / 1000.0);
    float speedKmph = speed * 3.6;

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Speed:");
    lcd.setCursor(0, 1);
    lcd.print(speedKmph);
    lcd.print(" km/h");

    if (speedKmph > 50) {
      digitalWrite(buzzer, HIGH);
      delay(5000);   // Buzzer ON for 5 sec
      digitalWrite(buzzer, LOW);
    }
    else {
      delay(3000);
    }
```

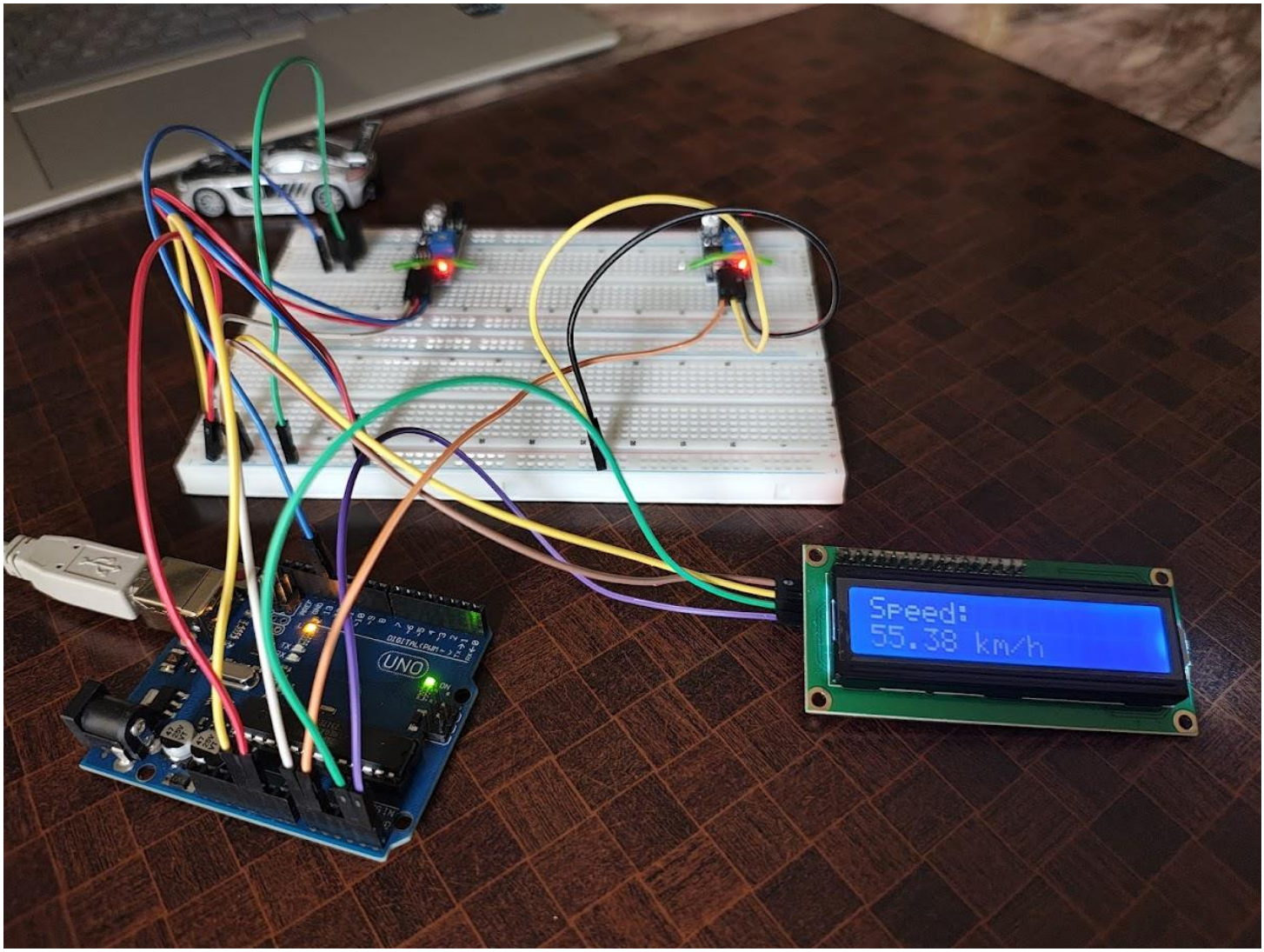
iot_pbl.ino

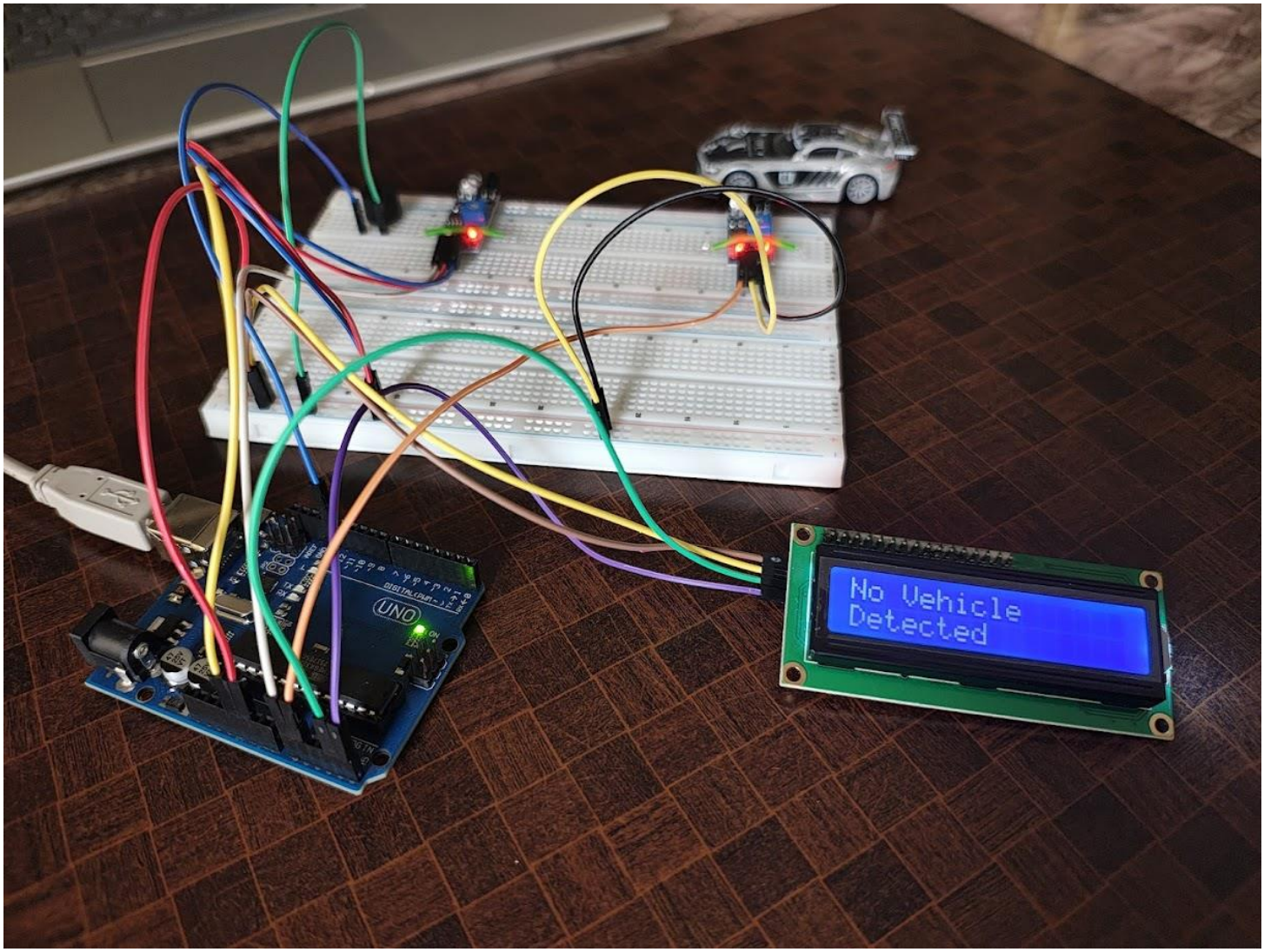
```
    lcd.setCursor(0, 0);
    lcd.print("Speed:");
    lcd.setCursor(0, 1);
    lcd.print(speedKmph);
    lcd.print(" km/h");

    if (speedKmph > 50) {
      digitalWrite(buzzer, HIGH);
      delay(5000);   // Buzzer ON for 5 sec
      digitalWrite(buzzer, LOW);
    }
    else {
      delay(3000);
    }

    lcd.clear();
  }

  firstSensor = 0;   // Reset
}

void showNoVehicle() {

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("No Vehicle");
  lcd.setCursor(0, 1);
  lcd.print("Detected");
  delay(2000);
  lcd.clear();

  firstSensor = 0;   // Reset
}
```

4.3 Sample Output




Speed:
40.00 km/h
UNO


Speed:
55.38 km/h
UNO


No Vehicle
Detected
UNO

5. Conclusion

The **Speed Detector System using Arduino and IR Sensors** successfully demonstrates how embedded systems and sensor technology can be used to measure the speed of a moving vehicle. In this project, two infrared sensors are used to detect the movement of a vehicle at two different points. The Arduino microcontroller measures the time taken for the vehicle to travel between the sensors and calculates the speed using the distance–time relationship.

The calculated speed is displayed on a 16×2 LCD display using an I2C module, making the output easy to read. In addition, a buzzer is used to provide an alert when the detected speed exceeds a predefined limit, indicating overspeeding. The system also handles cases where a vehicle is detected by only one sensor by displaying a "No Vehicle Detected" message.

This project provides a practical understanding of microcontrollers, sensors, and real-time data processing. It demonstrates how simple electronic components can be integrated to build an efficient speed monitoring system. The project can also serve as a basic model for traffic monitoring and speed control applications.

6. References

The following resources were referred to during the development of this project:

i. Arduino Official Website
https://www.arduino.cc

ii. Arduino Documentation and Tutorials
https://docs.arduino.cc

iii. Arduino LiquidCrystal I2C Library Documentation

iv. IR Sensor Module Datasheet and Working Principle

v. Online Arduino Programming Guides and Embedded Systems Tutorials

vi. YouTube Tutorials and Electronics Project Demonstrations related to Arduino Speed Detection Systems.